

Online Store Orders

1.Executive Summary

This document provides complete project documentation for the Online Store Orders data analysis and Power BI dashboard project. The initiative covers data acquisition, cleaning, exploratory analysis, and the development of an interactive business intelligence dashboard to support decision-making for an e-commerce operation.

2.Project Overview

2.1 Objectives

- Consolidate raw online store order records into a structured, analysis-ready dataset.
- Perform data cleaning and quality assurance to ensure accuracy and consistency.
- Conduct exploratory data analysis (EDA) to surface business insights.
- Build an interactive Power BI dashboard for stakeholder reporting.
- Document the full data pipeline, methodology, and findings for future reference.

2.2 Scope

The project covers order transactions recorded between 1 January 2023 and 30 June 2025. The dataset encompasses 1,200 individual orders from 1,189 unique customers, spanning 7 product categories and 5 order lifecycle statuses.

3.Data Description

3.1 Source Data

The source data is a single-sheet Excel workbook transactional records from an online retail store. Each row represents one order.

3.2 Data Schema

Field Name	Data Type	Description	Example
OrderID	String	Unique order identifier	ORD200000
Date	DateTime	Date the order was placed	2023-01-04
CustomerID	String	Unique customer identifier	C72649
Product	String	Product category purchased	Monitor
Quantity	Integer	Number of units ordered (1–5)	5
UnitPrice	Decimal	Price per unit (USD)	\$570.62

ShippingAddress	String	Street address for delivery	928 Main St
PaymentMethod	String	Mode of payment used	Debit Card
OrderStatus	String	Current fulfilment status	Shipped
TrackingNumber	String	Shipment tracking reference	TRK37947903
ItemsInCart	Integer	Total items in cart session (1–10)	7
CouponCode	String	Discount coupon applied	SAVE10
ReferralSource	String	Marketing channel that drove the order	Instagram
TotalPrice	Decimal	Total order value (USD)	\$2,853.10

3.3 Dataset Dimensions

Total Records	1,200 orders
Columns	14 fields per record
Date Range	1 January 2023 – 30 June 2025 (approx. 2.5 years)
Unique Customers	1,189 (near 1:1 customer-to-order ratio)
File Format	Microsoft Excel (.xlsx), single worksheet (Sheet2)

4.Data Cleaning & Preparation

4.1 Cleaning Steps Applied

The following data quality operations were performed prior to analysis:

4.1.1 Structural Integrity

- Verified row count consistency: 1,200 records with no missing rows.
- Confirmed no duplicate OrderIDs exist in the dataset.
- Standardised column headers to PascalCase with no special characters.

4.1.2 Data Type Enforcement

- Date column parsed and cast to datetime64 format.
- Quantity and ItemsInCart enforced as integer types.
- UnitPrice and TotalPrice enforced as 64-bit float (decimal currency).

- All categorical fields (Product, PaymentMethod, OrderStatus, CouponCode, ReferralSource) verified as strings.

4.1.3 Value Standardisation

- Categorical fields audited for inconsistent casing or whitespace — none detected.
- CouponCode and ReferralSource values validated against defined enumeration sets.
- TotalPrice cross-checked against $\text{Quantity} \times \text{UnitPrice}$ for logical consistency.

5.Key Insights & Recommendations

5.1 High Cancellation & Return Rate

At 41.4% combined, the cancellation and return rate is significant and warrants investigation. Recommended actions:

- Analyse cancellations and returns by product to identify high-risk categories.
- Survey customers post-cancellation to gather qualitative feedback.
- Review product descriptions and images to reduce expectation mismatches.
- Consider return policy adjustments for lower-risk products.

5.2 Instagram as Top Channel

Instagram drives the most orders (21.6%). Recommended actions:

- Increase advertising spend and creative investment in Instagram campaigns.
- Implement UTM tracking at SKU level to understand which products convert best via Instagram.
- Explore Instagram Shopping integration for a seamless purchase journey.

5.3 Diversified Payment Mix

The near-equal split across five payment methods suggests the business has already optimised for payment flexibility. Recommended actions:

- Ensure all payment gateways remain active and up to date.
- Monitor Gift Card usage for potential fraud signals.

5.4 Coupon Code Strategy

Three coupons are driving roughly equal usage. Recommended actions:

- Track revenue impact per coupon code to measure net margin contribution.
- Test time-limited or product-specific coupon codes to drive targeted demand.
- Evaluate whether FREESHIP increases basket size or reduces margin.

5.5 Laptop & Chair AOV Leadership

Laptops (\$1,110.56 AOV) and Chairs (\$1,098.99 AOV) generate the highest per-order revenue. Recommended actions:

- Prioritise these categories in paid media and email campaigns.
- Develop upsell and cross-sell strategies (e.g., Desk with Chair bundle).
- Review whether higher-AOV items are also more prone to returns or cancellations.

6. Conclusion

The Online Store Orders Data Analysis and Power BI Dashboard project successfully converted raw e-commerce transaction data into meaningful business insights and interactive visual reports. Through data cleaning, validation, exploratory analysis, and dashboard development, the project provided a clear understanding of sales performance, customer behaviour, marketing effectiveness, and order management trends.

7. Dashboard

